

What Did Context-PCG Learn?

Softmax-kernel in-context Bayesian near-field linear sampling

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The question: can the prompt teach us how to condition?

Bayesian near-field LSM requires two coupled solves

$$\begin{aligned} H_D Q_\mu &= \Phi, & H_D Q_\Sigma &= N_D K_\Gamma. \\ M_D &= K_\Gamma N_D^* Q_\mu, & \Sigma_D &= K_\Gamma - K_\Gamma N_D^* Q_\Sigma. \end{aligned}$$

- ▶ Original two-dimensional near-field acquisition.
- ▶ Deterministic point sources and receivers; one to six sound-soft obstacles.
- ▶ Mean accuracy alone is insufficient: the covariance solve controls numerical UQ.

Severe ill-conditioning $\kappa(H_D)$ up to 2.7×10^4

Matched budget $T = 32$ operator applications

Core comparison context-PCG vs CG

Modelling choice

The softmax/von-Mises nonlinearity fixes the GP feature space; it is not a learned temperature.

The learned object: one SPD factor per prompt

$$C_\theta(D, \Gamma) = U_\Gamma \text{diag}[g_\theta(U_\Gamma^* H_D U_\Gamma)]^{1/2} U_\Gamma^*, \quad g_\theta > 0.$$

$$\tilde{H}_D = C_\theta^* H_D C_\theta, \quad q = C_\theta x.$$

- ▶ The encoder reads the diagonal, row- ℓ_1 , and row- ℓ_2 norms of the prompt Hessian in the fixed GP basis.
- ▶ Shared token maps and mean pooling allow variable context length.
- ▶ The factor is computed once and frozen for all iterations: this remains standard PCG, not flexible CG.

What is learned

Positive, task-dependent modal gains.

What is not learned

The softmax kernel, an obstacle image, the LSM indicator, or a spatial regularization map.

Architectural restriction

Modal gains cannot rotate obstacle-dependent eigendirections outside U_Γ .

What was learned? Four controls isolate the signal

Method	Information available	Physical gain over CG	Interpretation
population-PCG	Acquisition distribution only	1.58×	Shared geometry helps, but fails under strong joint shift ($0.42\times$).
context-PCG	Current task Hessian statistics	2.05×	The prompt contains useful task-specific conditioning signal.
angular-Jacobi PCG	Exact current diagonal in GP basis	1.70×	Most available signal is cheaply accessible analytically.
hybrid-PCG	Analytic base + bounded learned correction	1.69×	The residual network adds no robust signal in the fixed basis.

Main inference

Context-PCG learns finite-horizon, task-dependent spectral weighting beyond a population factor. It does *not* learn a universally superior preconditioner: a correctly specified analytic GP-basis factor remains the stronger wall-clock control.

Held-out simulated near-field tasks; CG claims use the common physical residual.

Scaling study: what was actually run

Dataset size 128 $\rightarrow$ 32,768

Trained models 90

Network width 32 $\rightarrow$ 512

Dataset checkpoints 810

Context size 8 $\rightarrow$ 48

Task evaluations 362,880

Depth 1 $\rightarrow$ 128

Frozen seeds 3

Matched learned solvers

population-PCG, context-PCG, hybrid-PCG, Richardson, heavy-ball, Chebyshev.

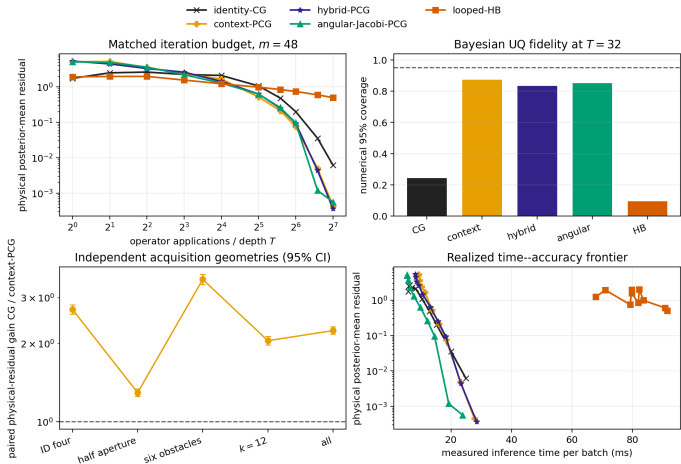
Training-free controls

CG, optimized CG, physical Jacobi, block-Jacobi, angular-Jacobi, exact dense solve.

Audits

Independent geometries, obstacle count, noise, aperture, frequency, joint shift, time, and UQ.

Direct answer versus ordinary CG



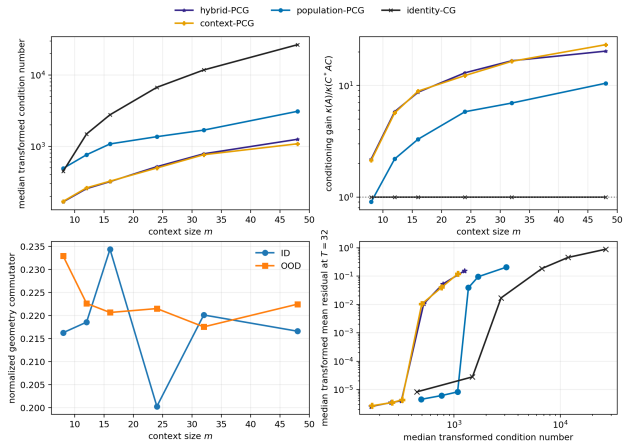
2.05 $\times$ physical-residual gain

6.44 $\times$ lower score error

Coverage: 0.243 $\rightarrow$ 0.873

Learned stationary loops remain inferior: even heavy-ball gives only 1.09 $\times$, AP 0.478, coverage 0.095, and 83.6 ms.

Conditioning mechanism: useful compression, incomplete basis



Raw median κ 2.66×10^4

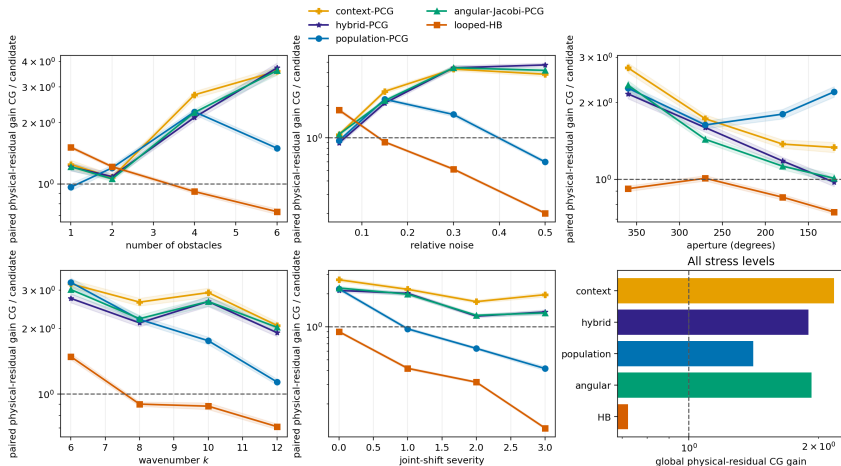
Context-PCG reduction $23.22 \times$

Angular-Jacobi reduction $24.28 \times$

What κ misses

Similar condition numbers can yield different finite-depth residuals: clustering and right-hand-side alignment also matter.

Robustness: prompt conditioning survives physical shifts



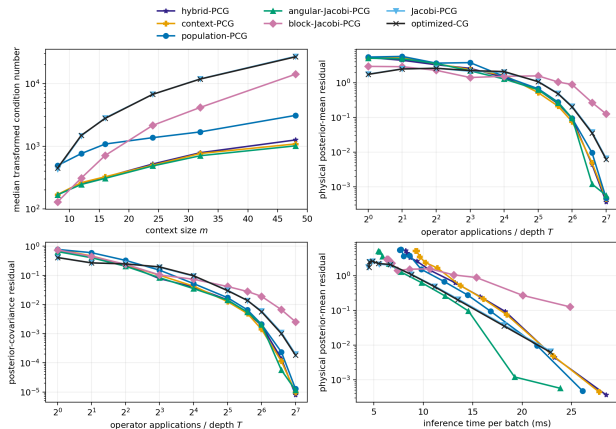
Gain range: $1.07 \times - 4.32 \times$

All context-PCG 95% lower
bounds > 1

Extreme joint shift: $1.96 \times$

Held-out simulated near-field tasks; CG claims use the common physical residual.

Learned versus informed analytic PCG



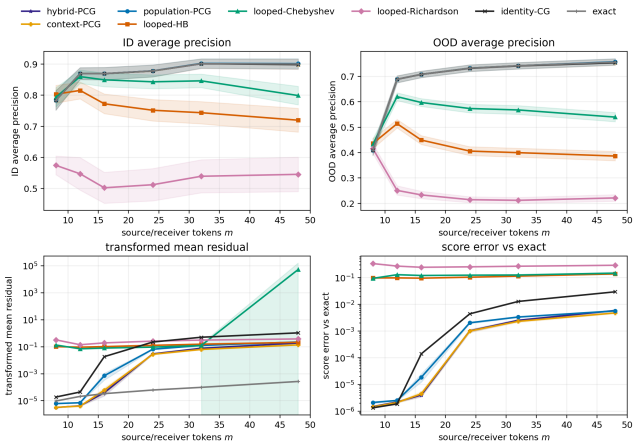
At $m = 48$, $T = 32$

- ▶ context-PCG residual: **0.521**
- ▶ angular-Jacobi residual: **0.629**
- ▶ angular-Jacobi is faster: **9.84 ms vs 13.75 ms**

Correct interpretation

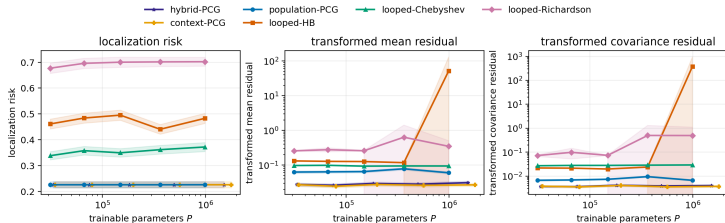
Context-PCG wins at fixed depth. Angular-Jacobi is faster and overtakes at high accuracy, reaching residual 0.01 in 19.24 ms.

Context scaling: solver fidelity improves, resolution saturates



Separation of effects. CG and PCG reach nearly the same localization AP; context-PCG mainly reduces numerical score error.

A bigger foundation model is not the answer



Best context width 64

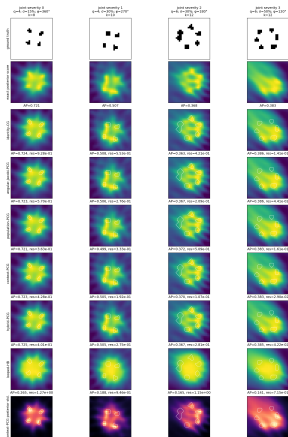
Largest width 512

Parameter ratio 22.45×

Observed plateau

Width 512 is 3.5% worse than width 64. Capacity is not the bottleneck; the fixed modal basis is.

Bayesian UQ: numerical fidelity, not universal calibration



What improves

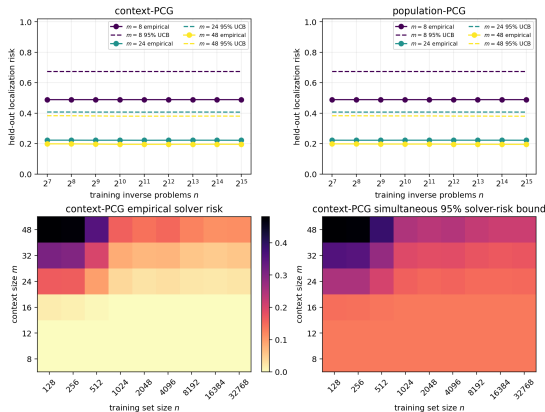
- ▶ Mean and covariance solves approach the exact decoder.
- ▶ Numerical 95% band coverage: $0.243 \rightarrow 0.873$ centrally.

What does not

- ▶ Independent-geometry AP improves by only 0.0021.
- ▶ Extreme-shift UQ coverage gain collapses to zero.

This is numerical posterior fidelity, not frequentist obstacle coverage.

Generalization evidence and its scope



Certified quantity

Simultaneous 95% held-out Hoeffding bounds over pre-specified settings.

Correct sample size

Held-out task count sets the slack; training size only indexes the learned hypothesis.

Scope

Only for the sampled simulator distribution—not arbitrary geometries, experiments, or random-source LSM.

Conclusions: what context-PCG learned

1. **Prompt-specific spectral weighting:** the current Hessian contains conditioning signal beyond shared acquisition geometry.
2. **Finite-horizon adaptation:** learned gains improve clustering/RHS alignment beyond what scalar condition number predicts.
3. **Robustness to unknown conditioning:** the gain over CG persists across every audited physical stress level.
4. **Numerical Bayesian fidelity:** both mean and covariance solves improve, while physical localization remains acquisition-limited.

Current limitation

A diagonal controller in a fixed GP basis cannot rotate task-dependent eigendirections.

Next architecture

Learn block-Lanczos or polynomial rotations from RHS-weighted Ritz moments; preserve exact outer Krylov coefficients.

One-line verdict

Clearly better than ordinary CG; not yet universally better than informed analytic PCG.